

Deployment for NOMA-UAV Base Stations Based on Hybrid Sparrow Search Algorithm

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Auxiliary UAV base stations are expected to be the most promising schemes to fill the resource shortage and the blind spot of coverage of the ground base stations in the future. This article proposes a hybrid sparrow search algorithm (SSA) to optimize the three-dimensional (3D) deployment of multi-UAV base stations with the objective of maximizing the sum log-rate utility. First, the air-to-ground channel model is established using the nonorthogonal multiple access technology to improve the spectrum utilization by considering the complex interference forms in multi-UAV scenarios, which is more compatible with the actual applications. Then, the multi-UAV deployment problem is described as an optimization problem to maximize the sum log-rate utility of all users. Finally, a hybrid SSA is proposed to implement the optimal 3D deployment of multi-UAV base stations. Specifically, in order to overcome the local optimization defects of the traditional SSA, the opposition-based learning strategy is utilized to modify the initial position of the sparrows to promote particle diversity, and the

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sine-cosine algorithm is developed to optimize the update formula of the beggar sparrows by integrating the disturbance into the update process. Consequently, a hybrid SSA with higher search accuracy, stability, and robustness is formed. The simulation results demonstrate that the proposed method outperforms the traditional methods in terms of the sum log-rate utility and throughput.

I. INTRODUCTION

Since the deployment of terrestrial wireless networks is usually planned based on long-term traffic behavior, the capacity and the demands cannot always match well, conflicting with the service requirements of users in some areas. This phenomenon will lead to poor resource allocation and underutilization. As a result, various resources sit idle at some stations, while some fluctuating demands cannot be satisfied at others. In order to avoid such drawbacks of the existing wireless networks, UAV base stations can be imported to locate the excess capacity requirements occurring in the networks. In this way, the network resources can be fully utilized, and users' service quality can be significantly improved [1], [2].

UAV-assisted base stations are almost the most feasible choice to provide signal coverage for remote areas, disaster zones, and battlefields. As flying base stations, UAVs can offer a promising approach to solving various problems and challenges in future mobile networks [3], [4]. The main differences between the UAV base stations and the ground base stations are as follows: the ground base stations are fixed with a determined communication range, high equipment complexity, and more mature technology. On the contrary, the flexibility of the UAV base stations can make their deployment more convenient. Furthermore, due to their high mobility, UAV flying base stations have unique features compared with the traditional static base stations, such as low risk, low cost, ease of operation, and responsiveness, which makes them move freely in the communication networks and adapt to network topology changes [5].

Although the UAV base stations may bring high benefits to the next generation of wireless networks, they also face many challenges. For instance, the UAV base stations have limited load and transmission power due to the current development of battery technology [6]. Besides, the signal of UAVs will experience longer line-of-sight (LoS) signal propagation when increasing the height, as shown in Fig. 1, and be affected both by LoS and non-line-of-sight (NLoS) links. Therefore, UAV base stations are more vulnerable to interference than ground communication. Additionally, the topology changes caused by the mobility of UAVs will lead to frequent intermittent connections and even transmission failures [7]. Fortunately, the optimized 3-D layout of UAV base stations can solve the above problems to some extent because the location of the UAV base station is one of the main factors affecting its battery life, base station coverage, signal transmission performance, and available capacity. The height of the UAV base stations directly affects the size of their coverage areas. Therefore, further studies should be performed on the quick deployment of the small and flexible UAV base stations in the 3D space to meet the

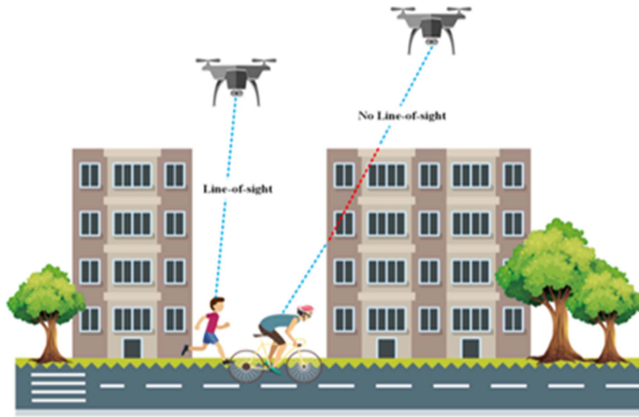


Fig. 1. LoS and NLoS links between UAV base stations and users.

communication coverage demand and maximize the network throughput performance.

On the other hand, with the popularization of smart terminals and the continuous emergence of new services such as virtual reality, the spectrum of resources cannot meet the exponentially growing requirements of mobile communication applications. In this case, nonorthogonal multiple access (NOMA) appeared in time [8], [9], [10], [11]. With serial interference cancellation (SIC) technology, NOMA can fulfill the spectrum resource constraints by increasing the complexity of the receivers [12]. In the traditional OFDM technology, subchannels are orthogonal and do not interfere with each other. In NOMA, a subchannel is shared by multiple users, and different users on the same subchannel transmit nonorthogonally. The NOMA transmitter employs power multiplexing technology for users on the same subchannel and allocates the signal power of different users according to the relevant algorithm. Accordingly, the receiver receives different signal powers for different users. The receiver adopts the SIC technology for multiuser detection to solve the interference between users and eliminates interference according to the power level of different users to realize correct demodulation [13]. In other words, the basic idea of NOMA transmission is to use the complex receiver design in exchange for higher spectral efficiency. This process can distinguish users and does not rely on the channel state information feedback, which improves the performance of the NOMA technology in high-speed mobile scenarios and provides superior mobile node backhaul links [14]. Therefore, this article introduces NOMA into the UAV auxiliary communication system.

II. RELATED WORK

The deployment of UAV auxiliary base stations has been widely concerned. In [15], an adaptive deployment scheme for the UAV-aided communication networks was proposed. The UAV serves the instantaneous traffic of random mobile users in the target unit by adjusting its displacement direction and distance. Literature [16] proposed a 3D deployment scheme for the UAV based on the heuristic algorithm, which ensures the connectivity of the UAV networks in static

and dynamic user scenarios. In [17], an intelligent UAV deployment scheme was discussed. In order to determine the energy-saving 3D position of UAVs with limited capacity without interference, three different layout algorithms were provided in the initial deployment: anticlockwise spiral algorithm, clockwise spiral algorithm, and hexagonal circle layout algorithm. After deployment, a Q-learning algorithm was proposed to adjust the existing UAV to maintain the downtime.

However, none of the previous articles mentioned the application of NOMA technology in the UAV auxiliary networks. In the traditional OFDMA system, the subcarriers are orthogonal, and the spectrum resources cannot be fully utilized. Due to its ability to multiplex multiple users with different power levels on the same frequency resource block, NOMA can further improve the spectral efficiency and throughput of the entire communication system. Actually, NOMA has been employed into some research works to improve the performance of UAV communication systems. In [18], an uplink NOMA network supported by UAV was developed, in which UAV is deployed to collect the information transmitted by the ground users. In order to maximize the user sum rate, the original problem of joint optimization of UAV deployment and power control under the transmission power and the quality-of-service constraints was proposed. Literature [19] focused on the NOMA power allocation, user pairing, and UAV deployment of UAV wireless systems. A heuristic pairing algorithm was proposed based on the sum of squares of minimum distances, and its pairing results were utilized for optimal power allocation and UAV layout. In [20], a UAV-aided transmission scheme of NOMA was proposed. The nonconvex problem was decomposed into two subproblems: power allocation and trajectory optimization, which are solved respectively.

However, the above documents only study the deployment of a single UAV. Due to the limited capacity and coverage, a single UAV cannot cover a large area. Multiple UAVs are often required to complete the network coverage in actual scenarios. The impact of the system interference should be considered when deploying multiple UAVs. Furthermore, multiple UAVs in NOMA should reuse the same spectrum, inevitably generating serious inter-cell interference. Therefore, it is necessary to consider not only inter-user interference but also the interference of other UAV base stations on the same layer to the target users. Although literature [21] employed the K-means clustering method to optimize the deployment of UAVs in the multi-UAV NOMA system, like many other swarm intelligence algorithms, the K-means clustering algorithm could not deal with the local optimization problem well. Hence, based on the above analyses, this article proposes a hybrid sparrow search algorithm (newSSA) for multiple UAV auxiliary base stations in the NOMA context, which takes the sum log-rate utility as the optimization goal to optimize the location deployment of UAV auxiliary base stations. The whole scheme considers the interference of other users within the coverage to the target users and the mutual interference

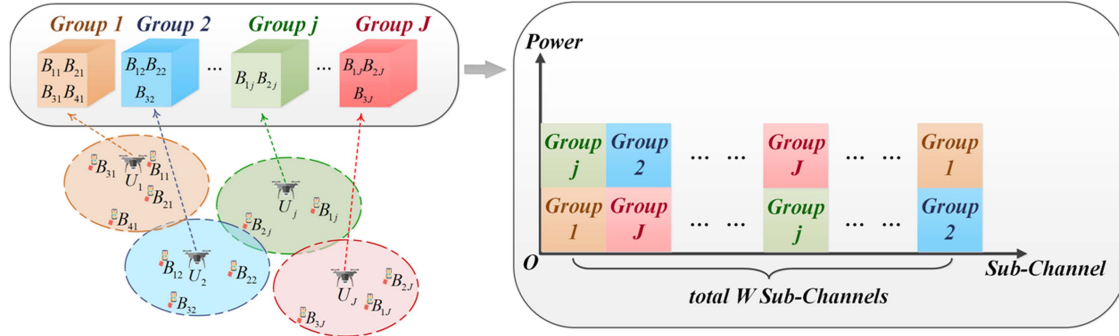


Fig. 2. Downlink framework of the NOMA-UAV network.

from other UAVs. Specifically, the main contributions of this article are reflected in the following four aspects.

- 1) NOMA technology is introduced into the UAV auxiliary communication system to establish a new air-to-ground channel model. Thus, each subchannel of the UAV base stations can be allocated to multiple users, and one user can receive signals from the base station through multiple subchannels. The UAV base stations can serve more users and improve spectrum utilization.
- 2) This article considers a realistic multi-UAV coverage scenario to overcome the limited coverage area of a single UAV base station. Instead of simply stacking the number of UAV base stations, we try to focus simultaneously on intra-group and inter-group interference. With these relatively complex interference assumptions that are more compatible with the actual application, multiple UAVs are restrained from each other and can obtain better deployment results to a certain extent.
- 3) This article introduces the new swarm intelligence algorithm, SSA, for the first time to solve the deployment problem of multi-UAV auxiliary base stations. Compared with other swarm intelligence algorithms, SSA has the characteristics of higher search accuracy and stronger robustness. UAV base stations and the ground users are mapped to the location of sparrows and food. Then, the optimized UAV location is found through a continuous iterative update.
- 4) In order to avoid the local optimization problem of the traditional SSA, a hybrid SSA is proposed. First, the opposition-based learning (OBL) idea is introduced to increase the disturbance to the initial position of UAVs. Moreover, the sine-cosine algorithm is developed to enhance the particle disturbance to the beggars in the position update process to reduce the dependence of SSA on the initial position and increase the optimization search speed.

III. SYSTEM MODEL

Fig. 2 shows a downlink auxiliary NOMA-UAV network. Suppose there are M users in the entire area,

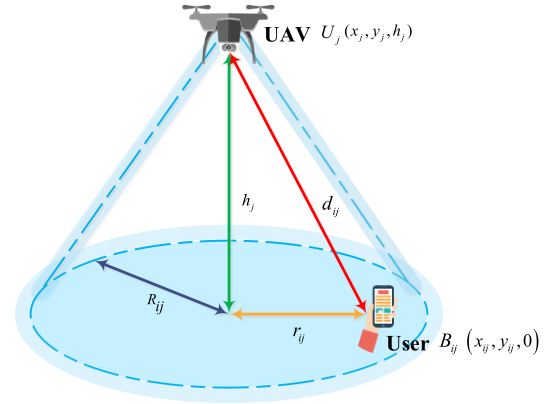


Fig. 3. Geometric diagram between UAV base station and user.

which is covered by J UAV base stations, denoted by the set of $\mathbf{U} = \{U_1, U_2, \dots, U_j, \dots, U_J\}$, ($j = \{1, 2, \dots, J\}$). Each UAV base station U_j sends signals to a group of ground users represented by $\mathbf{B}_j = \{B_{1j}, B_{2j}, \dots, B_{ij}, \dots, B_{Ij}\}$, ($i = \{1, 2, \dots, I\}$), where B_{ij} stands for the index of the i th user served by U_j . In other words, these M users are divided into J groups, with I users in each group. $j = \{1, 2, \dots, J\}$ is the group number set, and $i = \{1, 2, \dots, I\}$ is the set of user serial numbers in the group. Each user is allowed to access only one UAV base station simultaneously. The available bandwidth is divided into W subchannels, represented by the set $\mathbf{W} = \{1, 2, \dots, w, \dots, W\}$. Besides, according to the NOMA protocol, each subchannel of the UAV base station can be allocated to multiple users, and one user can receive signals from the base station through multiple subchannels. It is worth noting that ground users can only associate with the UAV base stations, excluding any ground base stations.

A. Air-to-Ground Channel Model

As shown in Fig. 3, taking the UAV's ground projection as the center, the coverage area of U_j is defined as a circular area with the center point as the coordinate origin. Let (x_j, y_j) be the projected position in the two-dimensional (2D) plane, namely the 2D position, and h_j be the altitude of the UAV base station U_j . Therefore, the 3D position coordinate of U_j is indicated by (x_j, y_j, h_j) . Suppose the ground

users do not move in the network. Thus, their locations are fixed, and the 3D position coordinate of the user B_{ij} can be expressed as $(x_{ij}, y_{ij}, 0)$. It is conceivable that the probability of successfully establishing the communication links has a direct relationship with (x_j, y_j, h_j) and $(x_{ij}, y_{ij}, 0)$.

Since LoS and NLoS links mainly affect the received signals of users from the UAV base stations just as described in the introduction, the air-to-ground channel suggested by the International Telecommunication Union [22] is employed to represent the channel model. In the LoS conditions, the wireless signal propagates along a straight line between the transmitter and receiver terminals without being blocked. In the NLoS conditions, wireless signals can only reach the receiver through reflection, scattering, and diffraction due to obstacles. Now, with the altitude h_j of U_j and the distance $r_{ij} = \sqrt{(x_{ij} - x_j)^2 + (y_{ij} - y_j)^2}$ between the ground user B_{ij} and the 2D position of U_j , the probability of LoS is modeled as

$$P_{LoS} = \prod_{q=0}^Q \left[1 - \exp \left(- \frac{\left[h_j - \frac{(q+\frac{1}{2})(h_j - h_{B_{ij}})}{Q+1} \right]^2}{2\mu_3^2} \right) \right] \quad (1)$$

$j = \{1, 2, \dots, J\}, i = \{1, 2, \dots, I\}$

where $Q = \text{floor}(r_{ij}\sqrt{\mu_1\mu_2} - 1)$ and $\text{floor}(\cdot)$ is a rounding down function. $h_{B_{ij}}$ is the altitude of B_{ij} . This probability is dependent on three statistical parameters related to the urban environment:

Parameter μ_1 : The ratio of built-up land area to the total land area.

Parameter μ_2 : The mean number of buildings per unit area (buildings/km²).

Parameter μ_3 : A scale parameter that describes the buildings' height distribution according to the Rayleigh probability density function (PDF), $f(H) = (H/\mu_3^2) \exp(-H^2/2\mu_3^2)$, where H is the building height in meters.

In order to further reduce this model, literature [23] fitted it with the real data of parameter μ_1 , μ_2 , and μ_3 from four urban, and obtained the following approximate expression:

$$P_{LoS} = \frac{1}{1 + a \exp \left(-b \left(\arctan \frac{h_j}{r_{ij}} - a \right) \right)}, \quad (2)$$

$j = \{1, 2, \dots, J\}, i = \{1, 2, \dots, I\}$

where a and b are directly related to μ_1 , μ_2 , and μ_3 . Thus, the probability of NLoS is

$$P_{NLoS} = 1 - P_{LoS}. \quad (3)$$

The path loss model for LoS and NLoS links in dB is described as

$$L_{LoS} = 20 \log \left(\frac{4\pi f_c d_{ij}}{c} \right) + \eta_{LoS} \quad (4)$$

$$L_{NLoS} = 20 \log \left(\frac{4\pi f_c d_{ij}}{c} \right) + \eta_{NLoS} \quad (5)$$

where $20 \log \left(\frac{4\pi f_c d_{ij}}{c} \right)$ represents the free space loss between U_j and B_{ij} . f_c is the carrier frequency, c is the speed of light, and d_{ij} is the distance between U_j and B_{ij} , defined as $d_{ij} = \sqrt{h_j^2 + r_{ij}^2}$. η_{LoS} and η_{NLoS} are the additional path losses of LoS and NLoS links, such as shadow and diffraction losses. They obey normal distribution of different parameters in dB scale, which are $\eta_{LoS} \sim N(\mu_{LoS}, \sigma_{LoS}^2)$ and $\eta_{NLoS} \sim N(\mu_{NLoS}, \sigma_{NLoS}^2)$. Obviously, this extra path loss of NLoS link is higher than that of LoS link. This means that a higher LoS probability can reduce the average path loss. Thus, the average path loss under the LoS and NLoS conditions can be written as

$$PL_{ij} = P_{LoS} \times L_{LoS} + P_{NLoS} \times L_{NLoS} \\ = \frac{\eta_{LoS} - \eta_{NLoS}}{1 + a \exp(-b(\arctan \frac{h_j}{r_{ij}} - a)) + 10 \log(h_j^2 + d_{ij}^2) + A} \quad (6)$$

$j = \{1, 2, \dots, J\}, i = \{1, 2, \dots, I\}$

where $A = 20 \log f_c + 20 \log(\frac{4\pi}{c}) + \eta_{NLoS}$, and c is the speed of light.

B. Downlink NOMA-UAV Model

With the definition of the average path loss PL_{ij} , a composite channel model consisting both of the small-scale and large-scale fading is considered [24]. Suppose that the channel coefficient of user B_{ij} on the subchannel w of U_j is expressed as $\xi_{ij}^w = \frac{g_{ij}^w}{PL_{ij}}$. Where PL_{ij} is the large-scale fading, which increases rapidly with the increase of d_{ij} , according to (6). The small-scale fading is defined by Rician fading in order to better illustrate the LoS link between U_j and B_{ij} . g_{ij}^w accounts for the Rician channel gain, which can be regarded as the reciprocal of Rician channel fading. Without losing generality, the users in each group are sorted according to their relative distance from U_j , i.e., $d_{1j} < d_{2j}, \dots, < d_{ij}, \dots, < d_{Ij}$. Accordingly, the order of their channel coefficients on the w th subchannel is $\xi_{1j}^w > \xi_{2j}^w, \dots, > \xi_{ij}^w, \dots, > \xi_{Ij}^w$ since the small-scale fading, a random sequence, has little to do with d_{ij} . This means that users who are closer to U_j have better channel conditions. According to the power domain NOMA principle [25], a lower transmit power is assigned to the user with a better channel condition, while the user with a weaker channel condition is assigned more power. Therefore, the order of power allocation factors of users should be $\beta_{1j}^w < \beta_{2j}^w < \dots, < \beta_{ij}^w, \dots, < \beta_{Ij}^w$, where β_{ij}^w represents the channel allocation factor assigned by U_j to user B_{ij} on subchannel w and

$$\sum_{i=1}^I \beta_{ij}^w = 1 \quad (7)$$

i.e., the power allocated for B_{ij} is $\beta_{ij}^w P_j$, where P_j is the total transmit power of U_j . Assume that the UAV base station U_j moves to a fixed-point position to serve its mobile users, while its communication channel is also stationary. Based on the above channel gain analyses, the SIC technology is

employed to first decode the signal with the most power allocation directly and then decode the signals of other users in turn. In each decoding process, SIC eliminates the interference of users with less power allocation than its own and treats the users' signals with more power allocation than its own as noise. At this point, the signal-to-noise ratio for any user B_{ij} on subchannel w can be defined as

$$\begin{aligned} \text{SINR}(B_{ij}^w) &= \frac{\beta_{ij}^w P_j |\xi_{ij}^w|^2}{\sum_{k=1}^{i-1} \beta_{kj}^w P_j |\xi_{kj}^w|^2 + \sum_{j' \in U \setminus j} \sum_{l=1}^{I'} \beta_{lj'}^w P_{j'} |\xi_{lj'}^w|^2 + \sigma^2} \\ j &= \{1, 2, \dots, J\}, i = \{1, 2, \dots, I\} \\ &= \frac{\beta_{ij}^w P_j |\xi_{ij}^w|^2}{\Gamma_1 + \Gamma_2 + \sigma^2} \end{aligned} \quad (8)$$

where σ^2 denotes the additive white Gaussian noise power. $\Gamma_1 = \sum_{k=1}^{i-1} \beta_{kj}^w P_j |\xi_{kj}^w|^2$ is the interference from other users with more power allocation in the same group. $\Gamma_2 = \sum_{j' \in U \setminus j} \sum_{l=1}^{I'} \beta_{lj'}^w P_{j'} |\xi_{lj'}^w|^2$ is the interference from neighbor UAV base stations in other groups [25]. Where j' indicates the group number of the neighbor UAV $U_{j'}$ and I' refers to the number of signals whose signal strength is stronger than its own. Therefore, the rate of B_{ij} served by U_j on its subchannels w can be expressed as

$$R_{ij}^w = \log(1 + \text{SINR}(B_{ij}^w)) \quad (9)$$

where $\log$ is the logarithm to the base 2.

C. Problem Formulation

Let μ_{ij}^w be a binary variable with the value of 1 or 0, indicating whether the user B_{ij} is associated with the UAV base station U_j or not, respectively. Our objective is to maximize the sum log-rate utility of all users by optimizing the 3D deployment of the UAV base stations. Thus, the deployment problem of UAV base stations, i.e., the best location of the drone cluster based on the maximization of the overall network utility value, can be expressed as

$$\begin{aligned} \text{maximize} & \sum_{i=1}^I \sum_{w=1}^W \sum_{j=1}^J \mu_{ij}^w \log(R_{ij}^w) \\ \text{s.t.} \quad \text{C1} : & \sum_{j=1}^J \mu_{ij}^w = 1 \quad \forall i \quad \forall w \\ \text{C2} : & \sum_{i=1}^I B_{ij} = B_j \quad \forall j \\ \text{C3} : & \sum_{j=1}^J B_j = B \\ \text{C4} : & \mu_{ij}^w \in \{0, 1\} \quad \forall i \quad \forall w \quad \forall j \end{aligned} \quad (10)$$

where B_j and B represent the total number of users in group j and the whole network. Constraint C1 indicates that each user within group j can only associate with one UAV base station, ensuring that each user in group j associated with

U_j is not mixed with other UAV base stations; constraint C2 ensures that the total number of users associated with each drone is equal to the number of users B_j assigned to each drone, thereby ensuring that all users in the group are not missed; and constraint C3 ensures that all users in the served area are served.

Consider the following parameter:

$$a_{ij}^w = \log(\log(1 + \text{SINR}(B_{ij}^w))). \quad (11)$$

According to relations (9) and (11), the UAV association problem (10) can be simplified as follows:

$$\begin{aligned} \text{maximize} & \sum_{i=1}^I \sum_{w=1}^W \sum_{j=1}^J \mu_{ij}^w a_{ij}^w \\ \text{s.t.} \quad \text{C1} : & \sum_{j=1}^J \mu_{ij}^w = 1 \quad \forall i \quad \forall w \\ \text{C2} : & \sum_{i=1}^I B_{ij} = B_j \quad \forall j \\ \text{C3} : & \sum_{j=1}^J B_j = B \\ \text{C4} : & \mu_{ij}^w \in \{0, 1\} \quad \forall i \quad \forall w \quad \forall j. \end{aligned} \quad (12)$$

IV. PROPOSED APPROACH

This section proposes an improved SSA to effectively place the UAV base stations in 3D space.

A. Traditional SSA

SSA is a new swarm intelligence algorithm proposed in 2020 [29]. In SSA, sparrows are divided into producers, beggars, and forerunners. Every sparrow has a fitness value determined by the optimized function. The algorithm judges whether the current position of the sparrow is better than the previous position according to the fitness function. Producers have high energy reserves and fitness values, mainly providing foraging areas and directions for other sparrows. The beggars follow the producers to find food to save energy and increase their fitness value. Some beggars may also constantly monitor the producers and compete for food. The forerunners will release an alarm while facing dangers and quickly move to the safe area to obtain a better position. Assuming Z sparrows in an E -dimensional search space, the mathematical model and the location of each sparrow can be described as follows:

$$\Phi = \begin{bmatrix} \varphi_{1,1} & \varphi_{1,2} & \cdots & \varphi_{1,e} & \cdots & \varphi_{1,E} \\ \varphi_{2,1} & \varphi_{2,2} & \cdots & \varphi_{2,e} & \cdots & \varphi_{2,E} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \varphi_{z,1} & \varphi_{z,2} & \cdots & \varphi_{z,e} & \cdots & \varphi_{z,E} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \varphi_{Z,1} & \varphi_{Z,2} & \cdots & \varphi_{Z,e} & \cdots & \varphi_{Z,E} \end{bmatrix} \quad (13)$$

where Φ is the mathematical model matrix, and each element $\varphi_{z,e}$ represents the position of the z th sparrow in the e th dimension.

Since the producers guide the flow of the whole sparrow group and can find food anywhere, their location is updated as follows:

$$\varphi_{z,e}^{s+1} = \begin{cases} \varphi_{z,e}^s \exp\left(\frac{-s}{\alpha \cdot \text{Iter}_{\max}}\right) & R_2 < \text{ST} \\ \varphi_{z,e}^s + Q \cdot L & R_2 \geq \text{ST} \end{cases} \quad (14)$$

where s denotes the current number of iterations; $\text{Iter}_{\max}$ is the maximum number of iterations; α is a random number between 0 and 1; R_2 and ST denote the alert value and the safety threshold, respectively; Q is a random number obeying the normal distribution, and L is a matrix whose all elements are 1 [29].

The updated formula of beggars is as follows:

$$\varphi_{z,e}^{s+1} = \begin{cases} Q \cdot \exp\left(\frac{\varphi w_e^s - \varphi_{z,e}^s}{z^2}\right) & z > \frac{Z}{2} \\ \varphi b_e^{s+1} + \frac{1}{E} \sum_{e=1}^E (|\varphi_{z,e}^s - \varphi b_e^{s+1}| \text{rand}\{-1, 1\}) & z \leq \frac{Z}{2} \end{cases} \quad (15)$$

where φw_e^s and φb_e^s represent the global worst and best positions of the beggar sparrow in the e th dimension at the s th iteration. When $z > Z/2$, the z th participant does not get food and is hungry with low adaptability. In order to obtain higher energy, it should fly to other places to find food; otherwise, the z th beggar will randomly find a position near the best position of the producer to forage [29].

The updated formula of forerunners is as follows:

$$\varphi_{z,e}^{s+1} = \begin{cases} \varphi b_e^s + \gamma |\varphi_{z,e}^s - \varphi b_e^s| & \Omega_z \neq \Omega_b \\ \varphi_{z,e}^s + K \cdot \left(\frac{|\varphi_{z,e}^s - \varphi w_e^s|}{|\Omega_z - \Omega_w| + \varepsilon}\right) & \Omega_z = \Omega_b \end{cases} \quad (16)$$

where γ has a random normal distribution with a mean value of 0 and a variance of 1. As the step control parameter, K is a random number between [-11]; Ω_z , Ω_b , and Ω_w represent the current fitness value of the forerunner sparrow, the current global best fitness value, and the worst fitness value, respectively; ε is a minimum constant that prevents the denominator from being zero [29].

B. Improved NewSSA Solution

The specific scenario of this article assumes that each alternative location of U_j is a sparrow, forming a total of Z sparrows. Each sparrow is located in a 3D space, i.e., $E = 3$. Thus, the mathematical model in (13) is specifically mapped as

$$\Phi^j = \begin{bmatrix} \varphi_{1,1} & \varphi_{1,2} & \cdots & \varphi_{1,e} & \cdots & \varphi_{1,E} \\ \varphi_{2,1} & \varphi_{2,2} & \cdots & \varphi_{2,e} & \cdots & \varphi_{2,E} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \varphi_{z,1} & \varphi_{z,2} & \cdots & \varphi_{z,e} & \cdots & \varphi_{z,E} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \varphi_{Z,1} & \varphi_{Z,2} & \cdots & \varphi_{Z,e} & \cdots & \varphi_{Z,E} \end{bmatrix}_j$$

$$= \begin{bmatrix} x_1 & y_1 & h_1 \\ \vdots & \vdots & \vdots \\ x_z & y_z & h_z \\ \vdots & \vdots & \vdots \\ x_Z & y_Z & h_Z \end{bmatrix}_j. \quad (17)$$

Like other swarm intelligence algorithms, SSA also faces less population diversity and easily falls into local optimum. Therefore, before optimizing the position of UAV base stations, OBL strategy is introduced to improve the initialized position of the UAV base stations [26]. The central idea of the OBL strategy is to optimize the next iteration by finding the reverse solution that is better than the current solution and replacing it. Although the reverse solution obtained by the traditional OBL strategy works well in the early iterations, it can easily fall into the local optimal region in the middle and late iterations, weakening the optimization effect. Therefore, to solve the problem of the traditional OBL strategy to a certain extent, this section combines the OBL strategy with the refraction principle to increase the random disturbance of the UAV base stations and promote the population's diversity and global search ability [26].

The inverse solution resulting from the improved OBL strategy based on the light refraction principle can be expressed as

$$(x_z, y_z, h_z)^* = \left(\frac{x_{\max} + x_{\min}}{2} + \frac{x_{\max} + x_{\min}}{2kn} - \frac{x_z}{kn}, \frac{y_{\max} + y_{\min}}{2} + \frac{y_{\max} + y_{\min}}{2kn} - \frac{y_z}{kn}, \frac{h_{\max} + h_{\min}}{2} + \frac{h_{\max} + h_{\min}}{2kn} - \frac{h_z}{kn} \right) \quad (18)$$

where $(x_z, y_z, h_z)^*$ is the optimized initial position; $x_{\max}$, $x_{\min}$, $y_{\max}$, $y_{\min}$, $h_{\max}$, and $h_{\min}$ are the dynamic maximum and minimum of x_z , y_z , and h_z , respectively. This dynamic boundary setting is conducive to the algorithm's convergence to a better search space and accelerates the convergence speed. The two parameters k and n are the scaling and fine-tuning factors, which can increase the probability of the algorithm jumping out of the local optimal solution. By adjusting the value of k and n to increase the population diversity, the global search ability of the algorithm can be improved.

Like the traditional SSA algorithm, the location update formula for the producers is as follows:

$$(x_z^{s+1}, y_z^{s+1}, h_z^{s+1}) = \begin{cases} (x_z^s, y_z^s, h_z^s) \cdot \exp\left(\frac{-s}{\alpha \cdot \text{Iter}_{\max}}\right), & R_2 < \text{ST} \\ (x_z^s, y_z^s, h_z^s) + Q \cdot L, & R_2 \geq \text{ST} \end{cases} \quad (19)$$

when $R_2 < \text{ST}$, no dangers appear around this search area, and the producers can conduct a wide search of the area. When $R_2 \geq \text{ST}$, a producer has found danger in the search area, so all sparrows in that search area move to the next safe search area. In the scenario of this article, since there will be no real predators, the danger alert can be replaced

by the trend of falling into local optimum immediately. In other words, if the sparrow finds that the trend of falling into the local optimum is more obvious, it is considered a danger, and the position should be adjusted.

If the location searched by the producers is at the local optimal position, many beggars will follow the producers into these local optimal locations, which will cause the whole population to stop moving and cause the loss of population diversity. Therefore, to further reduce the probability of local optimization, the idea of a sine-cosine search algorithm is introduced to optimize the update formula of the beggars [27]. The sine-cosine search algorithm utilizes the advantages of sine and cosine functions in balancing the algorithm's global exploration and local development ability in the search process by adaptively changing the amplitude of the two functions. Finally, the global optimal solution can be found. The location update formula of beggars can be derived as where (xb_z^s, yb_z^s, hb_z^s) is the current global best position. r_1 , r_2 , r_3 , and r_4 are four main parameters, where r_1 is a control parameter, which is mainly responsible for controlling the amplitude of sine and cosine functions, defined by the following formula:

$$r_1 = \left(1 - \left(\frac{s}{\text{Iter}_{\max}}\right)^\eta\right)^{\frac{1}{\eta}} \quad (21)$$

where η is the adjustment factor, and $\eta > 1$. $r_2 \in [0, 2\pi]$ is a random number that determines the distance the sparrow moves; $r_3 \in [0, 2\pi]$ is a random number that controls the influence of the optimal individual on the sparrow's next position. r_4 equally switches between the sine and cosine components in (20) shown at the bottom of this page. Besides, the nonlinear weighting factor w , expressed by (22), is introduced to adjust the dependence of the position updates on the current position information

$$w = \frac{e^{\frac{s}{\text{Iter}_{\max}}} - 1}{e - 1} \quad (22)$$

where e is the natural constant. The changing trend of r_1 and w can be easily obtained from (21) and (22), as shown in Fig. 4 equation (23) shown at the bottom of next page.

The optimization algorithm assumes that the number of forerunners in danger is 15% of the total population and randomly selects the initial forerunner individuals in danger in the population. These sparrows quickly move to the safe area because they realize they are about to fall into local optimization. Therefore, the position of the forerunners is updated as follows:

The number of producers and forerunners of the sparrows is indicated by PR and FR , respectively. As a result, the specific algorithm steps are shown in Table I.

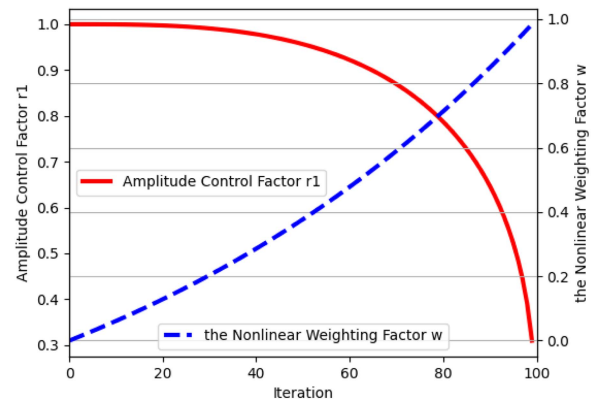


Fig. 4. Changing trend of r_1 and w .

V. SIMULATION RESULTS

In this section, simulations are performed to compare the performance of the proposed newSSA method with the existing genetic algorithm (GA) method [28] and the traditional SSA method [29]. The simulation parameters are shown in Table II.

The distribution of users significantly affects the key performance indicators of the network. Although the Poisson point process (PPP) model has been employed to describe the users' location in recent studies, the PPP model still cannot fully express the distribution characteristics of the users, such as clusters. Actually, the actual user distribution is rarely a pure PPP. Network users usually concentrate in residential or office buildings, shopping malls, bus stations, and other social places. It is like being attracted by them, thus forming a clustered structure. Therefore, the simulation employs a cluster model, called the Poisson cluster process or super-Poisson process [30], to describe the user distribution model. Thomas process [30] is chosen to generate the super-Poisson pattern. First, some buildings obeying PPP are selected as cluster heads in the network area. Then, users are located around the cluster heads. The users associated with each cluster are distributed using symmetric Gaussian distributed distance $N(0, \delta)$ from the cluster head and uniformly distributed in $[0, 2\pi]$ in the spheres centered at the cluster heads.

Fig. 5 shows the sum log-rate utility for GA, SSA, and newSSA for different iterations. It can be observed that the proposed newSSA method has the best sum log-rate utility value compared with the traditional SSA and GA methods. The main reason is that the traditional SSA and GA search processes both strongly depend on the initial population. Therefore, they easily fall into the "early maturity" phenomenon. Besides, since implementing the GA method needs many parameters, such as the crossover rate

$$(x_z^{s+1}, y_z^{s+1}, h_z^{s+1}) = \begin{cases} w(x_z^s, y_z^s, h_z^s) + r_1 \sin(r_2) |r_3(xb_z^s, yb_z^s, hb_z^s) - (x_z^s, y_z^s, h_z^s)| & r_4 < 0.5 \\ w(x_z^s, y_z^s, h_z^s) + r_1 \cos(r_2) |r_3(xb_z^s, yb_z^s, hb_z^s) - (x_z^s, y_z^s, h_z^s)| & r_4 \geq 0.5 \end{cases} \quad (20)$$

TABLE I
Algorithm Steps Based on the Improved SSA Algorithm (NewSSA)

Algorithm	newSSA 3D Placement Optimization
Input:	$z, PR, FR, Iter_{\max}$
Output:	$(xb_z^s, yb_z^s, hb_z^s), \Omega_b$
1:	Initialization
2:	The initial position of the ground user B_{ij} was randomly selected within the target area.
3:	Initialization of the sparrow population location.
4:	Update (x_z^s, y_z^s, h_z^s) with (18)
5:	while ($s < Iter_{\max}$)
6:	The fitness values of all sparrows in the current population are calculated and ranked, and the best individual with the highest fitness value and the worst individual with the lowest fitness value are selected.
7:	for $i = 1: PR$
8:	Update $(x_z^{s+1}, y_z^{s+1}, h_z^{s+1})$ with (19)
9:	end for
10:	for $i = (PR + 1): Z$
11:	Update $(x_z^{s+1}, y_z^{s+1}, h_z^{s+1})$ with (20)
12:	end for
13:	for $i = 1: FR$
14:	Update $(x_z^{s+1}, y_z^{s+1}, h_z^{s+1})$ with (23)
15:	end for
16:	Get the current best position and maximum fitness value.
17:	If the current position is better than the previous one, the optimal position is updated.
18:	$s = s + 1$
19:	end while
20:	return $(xb_z^s, yb_z^s, hb_z^s), \Omega_b$

and the variation rate, the choice of these parameters significantly affects the algorithm's reaction and search speed, and most of these parameters can only be chosen empirically. These problems degrade the performance of the GA method. Unlike these two algorithms, the proposed newSSA method adds the optimal setting of the initial position of sparrows with the OBL method to avoid falling into the local optimum, as shown in (18).

Fig. 6 describes the throughput for the three methods for different iterations. As shown in Fig. 6, the total throughput increases with the number of iterations for all three methods

TABLE II
Simulation Parameters

Parameter	Mean	Value
a	The environmental parameters	9.61
b	The environmental parameters	0.16
f_c	The carrier frequency	2 GHz
c	The speed of light	3.0×10^8
η_{LoS}	The average additional loss varying of LoS	1 dB
η_{LoS}	The average additional loss varying of NLoS	20 dB

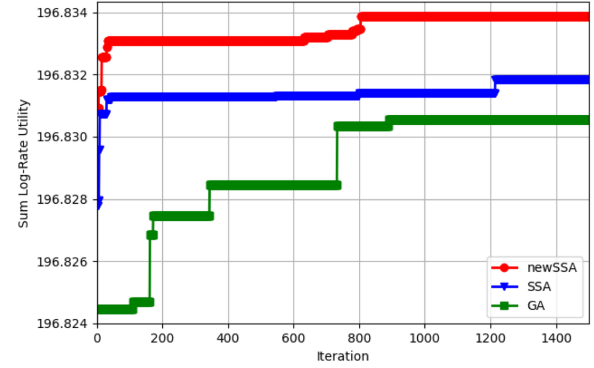


Fig. 5. User log ratio for GA, SSA, and newSSA versus iteration.

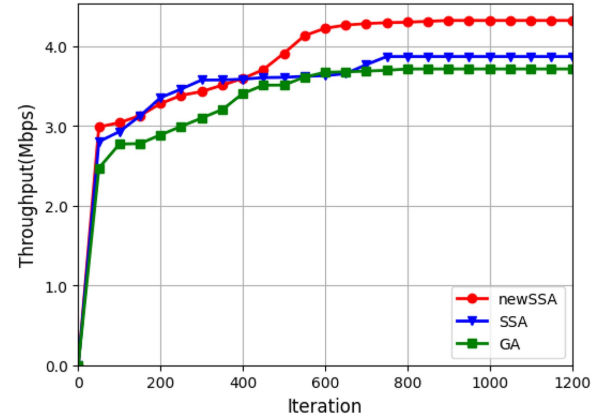


Fig. 6. Throughput for GA, SSA, and newSSA versus iteration.

at the beginning of the curves. This is intuitive because as the number of iterations increases, the UAV base stations reach better locations to serve the users. Then, as the number of iterations further increases, the throughput will not be further improved because the network has reached saturation. At the same time, it can be seen that the newSSA method can obtain the highest throughput performance among the three algorithms. Concretely, the advantage of the newSSA

$$(x_z^{s+1}, y_z^{s+1}, h_z^{s+1}) = \begin{cases} (xb_z^s, yb_z^s, hb_z^s) + \gamma |(x_z^s, y_z^s, h_z^s) - (xb_z^s, yb_z^s, hb_z^s)| & \Omega_z \neq \Omega_b \\ (x_z^s, y_z^s, h_z^s) + K \left(\frac{|(x_z^s, y_z^s, h_z^s) - (xw_z^s, yw_z^s, hw_z^s)|}{|\Omega_z - \Omega_w| + \varepsilon} \right) & \Omega_z = \Omega_b \end{cases} \quad (23)$$

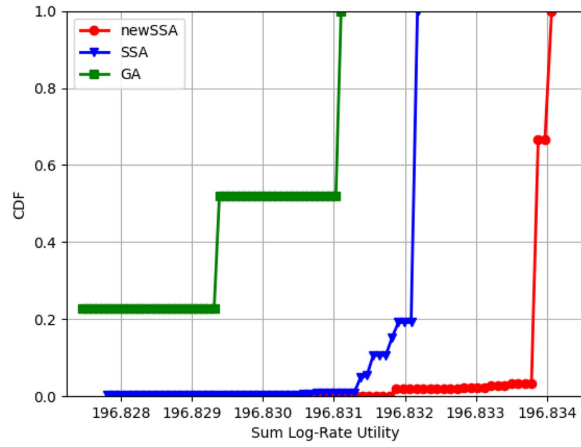


Fig. 7. CDF for GA, SSA, and newSSA versus fitness.

method is 11.26% over the original SSA in the simulation and 16.30% over the GA method after the convergence. This is because the newSSA method introduces the NOMA technology so that each subcarrier can access multiple users and improve spectrum utilization. Besides, this is also due to ordering users according to distance in the downlink NOMA-UAV model and the comprehensive investigation and processing of intra-group and inter-group interference in the (8).

Fig. 7 compares the cumulative distribution function (CDF) performance of the sum log-rate utility for users in the newSSA method with the other two methods. CDF is the integral of the PDF, which can completely describe the probability distribution of a random variable. The curve of CDF means taking a value of the sum log-rate utility and calculating the probability that the currently obtained sum log-rate utility is less than this value. SSA and GA's inflection point is when the sum log-rate utility reaches 196.832 and 196.830, the corresponding ordinates are 0.85 and 0.88, which means that the sum log-rate utility obtained by SSA and GA algorithms is 85% and 88% below 196.832 and 196.830, while the sum log-rate utility obtained by newSSA is less than 33% below 196.832. On the one hand, this improvement is due to the features of SSA, such as high search accuracy, fast convergence, good stability, and strong robustness compared with other swarm intelligence algorithms. On the other hand, the newSSA method can adjust the parameters, such as the amplitude of sine and cosine functions r_1 , and the moving distance parameter r_2 in real-time according to the change of the fitness value, which improves the throughput performance.

As mentioned above, the proposed newSSA method employs the sine-cosine search algorithm to improve the location update of beggars who occupy most of the sparrows. It employs the amplitude of the sine and cosine functions r_1 and a nonlinear weighting factor w to improve the algorithm's performance. In order to verify the impact of these two factors on the improvement of network performance on the newSSA method, four different combinations of these two factors are employed in the simulations, as shown in Table III. In Table III, $\circ$ and $\times$ indicate that the

TABLE III
Four Different Combinations of r_1 and w

	Amplitude control factor r_1	The nonlinear weighting factor w
newSSA	$\circ$	$\circ$
newSSA1	$\circ$	$\times$
newSSA2	$\times$	$\circ$
newSSA3	$\times$	$\times$

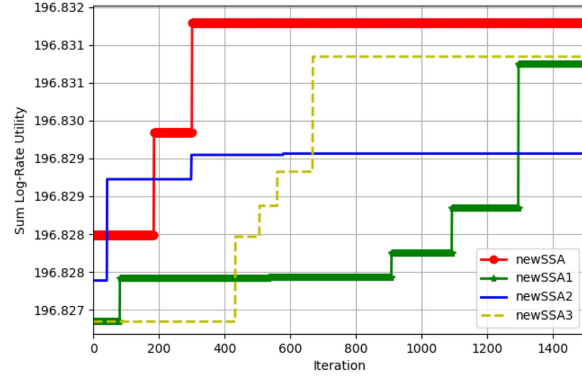


Fig. 8. Sum log-rate utility for different forms of newSSA versus iteration.

algorithm is implemented with and without the factor. It can be seen that newSSA applies both r_1 and w . NewSSA1 only applies the parameter r_1 , while newSSA2 only applies the parameter w . In contrast, newSSA3 employs none of these two parameters.

Fig. 8 describes the sum log-rate utility for newSSA, newSSA1, newSSA2, and newSSA3 versus iteration. When the two factors are added simultaneously, the best performance improvement of the newSSA method is obtained. These two parameters cooperate to achieve the best effect of the algorithm. First, the parameter w is incorporated into the updated (20) of beggars to increase the disturbance to the current position, and the size of w can be adjusted at different stages to achieve different purposes of increasing the global search capability or accelerating the convergence speed. Specifically, as shown in Fig. 4, w has a small value at the early stage of the iteration and its growth rate is not fast, while its value is significant at the later stage. The smaller w in the early stage reduces the dependence of individual position updates on the current solution position and improves the algorithm's global search ability. The larger w in the later stage can accelerate the algorithm's convergence rate considering the high dependence of the location update on the current location. Similarly, the larger r_1 value in the initial stage increases the disturbance to the position update using the sine-cosine function, thus enhancing the diversity of particles. At the same time, the gradual reduction of r_1 in the later stage improves the convergence speed. Besides, the newSSA3 curve performance without the two parameters is better than applying one parameter alone (newSSA1 and newSSA2), which indicates that the

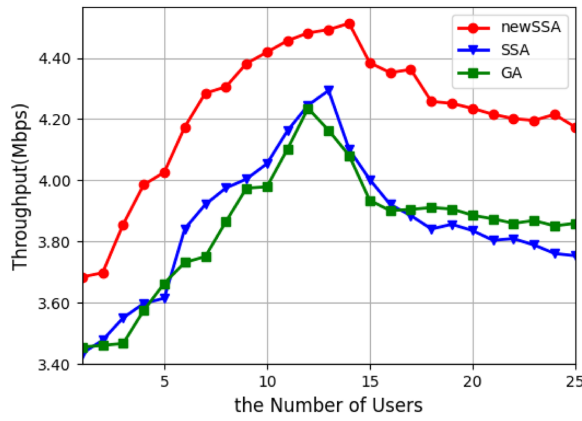


Fig. 9. Throughput for GA, SSA, and newSSA versus the number of users.

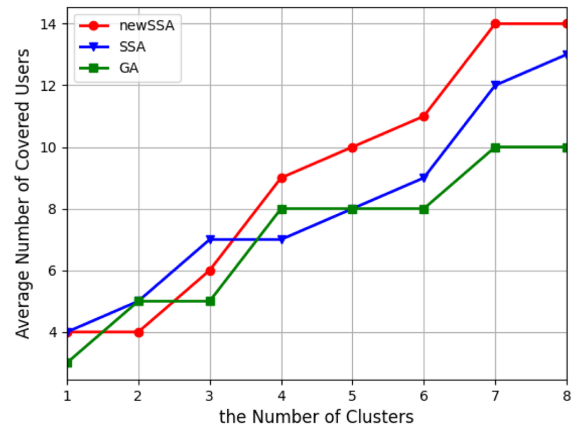


Fig. 11. Average number of covered users for GA, SSA, and new SSA versus the number of clusters.

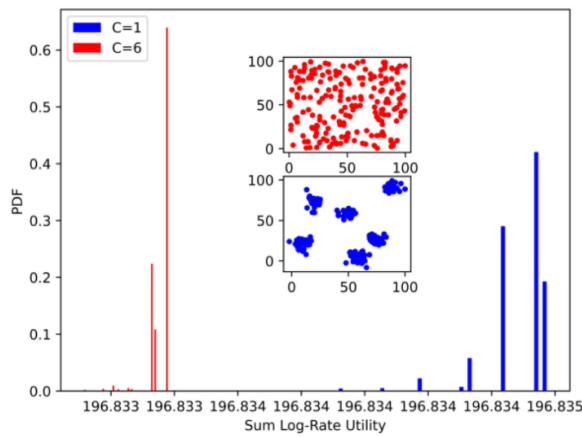


Fig. 10. PDF and exemplary distributions for $C=1$ and $C=6$.

two parameters should be employed simultaneously, and the algorithm's performance cannot be improved with only one parameter.

Fig. 9 depicts the throughput for GA, SSA, and newSSA versus the number of users. As shown in Fig. 9, the throughput of the newSSA method is significantly higher than the other two algorithms for the same number of users. In the highest part, the advantage of the newSSA method is 12.07% over the traditional SSA in the simulation and 14.22% over the GA method. This is mainly because the newSSA method considers both the interference between users in the same group and the inter-group interference caused by other UAV base stations, thus achieving higher throughput performance under the same number of users. Additionally, in the later stage of the three curves, the throughput performance of all three algorithms decreases because the channel collision probability increases with the number of users.

Fig. 10 shows the PDF of the sum log-rate utility of newSSA for users for $C=1$ and $C=6$, as well as the exemplary distributions of users corresponding to each value of C . It is assumed that C is the coefficient of variation

of the user distribution, which can describe the degree of aggregation and dispersion of the user distribution. The super-Poisson process has $C > 1$ and PPP has $C = 1$ [30]. It can be observed that each user has a larger probability of receiving a higher sum log-rate utility in a more clustered distribution. This confirms that the newSSA can improve the network's user satisfaction more effectively under a higher degree of cluster distribution. That is to say, compared with the completely independent distribution of users, the newSSA method is more suitable for the higher clustering of user distribution. This is because the cluster distribution of users is more conducive to the UAV base stations flying to favorable terrain for better coverage, and the newSSA method can handle the interference problem well. Therefore, by increasing the value of C , more users can be associated with the UAV base stations, thus achieving a better sum log-rate utility in the system.

Fig. 11 vividly describes the average number of covered users for GA, SSA, and newSSA versus the number of clusters. It can be seen that the number of users covered by the UAV base stations in newSSA does not exhibit advantages for a small value of C , but it gradually exceeds the other two algorithms as C increases. It is confirmed that the proposed newSSA method is more suitable for scenarios with more clustering distributions than the other two algorithms. This is mainly due to various avoidance measures for local optimization in newSSA, so the local optimum cannot occur even if users are distributed in clusters. This demonstrates the advantage of the newSSA method in the user coverage performance over the traditional methods.

VI. CONCLUSION

This article proposes a dynamic deployment optimization of multi-UAV base stations based on the hybrid SSA to provide wireless coverage. First, the considered on-demand placement problem is represented with a NOMA communication network model combined with the air-to-ground channel characteristics of the UAV networks. Combined with the assumption of the user's distance from the UAV

base stations and the processing ability of SIC technology for the interference, the complex interference in the scene is appropriately handled according to the power domain NOMA principle. Then, the hybrid SSA is introduced to solve the 3D deployment problem of multi-UAV base stations. That is, to avoid the so-called local optimization problem, the OBL and the sine-cosine algorithm are integrated into the traditional SSA to form a hybrid SSA with better performance. The specific functions of OBL and the sine-cosine algorithm are as follows: add disturbances to the initial particles and the update process of the beggars to increase the diversity of particles, reduce the dependence of the algorithm on the initial position, enhance the global search ability, and improve the convergence speed. Simulation results indicate that the proposed newSSA method outperforms the traditional SSA and GA methods in terms of the sum log-rate utility and throughput while considering the system's interference. In the future, we will improve the proposed model's construction, global search ability, and convergence speed to make the algorithm more universal. It is worth mentioning that, at present, the complexity of the receivers with SIC technology restricts the application of NOMA in practice. We expect that with the enhancement of chip processing capability, the application of NOMA in practical systems will become possible.

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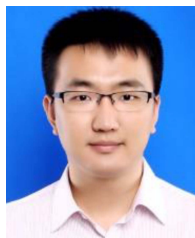
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